VELS INSTITUTE OF SCIENCE, TECHNOLOGY AND ADVANCED STUDIES

DEPARTMENT: INFORMATION TECHNOLOGY
SEMESTER: I
TITLE OF THE COURSE: DATA MINING AND DATA WAREHOUSING
COURSE CODE: 24DMIT12

TIME: 03 Hours **Max. Marks: 100**

SECTION – A (3 MARKS)

S. No	Questions	Unit	K-Level	CO	PO
1.	Define Data Mining, what are its various names.	I	1	1	PO1,PO2
2.	What is Data Cleaning?	I	1	1	PO1
3.	What are the alternative names of Data Mining?	I	1	1	PO1
4.	Why Data Mining is important?	I	1	1	PO1
5.	What is Transactional Database?	I	1	1	PO2
6.	Name the steps involved in Data Mining.	I	1	1	PO1
7.	Why do we preprocess the data?	I	1	1	PO2
8.	What is Data Transformation?	I	1	1	PO1
9.	What is Clustering?	I	1	1	PO1
10.	Define Association Rule Learning.	I	1	1	PO1
11.	Name some of the types of algorithms used in Association Rule Learning.	I	1	1	PO2
12.	Name the metrics to measure the association between data items.	I	1	1	PO2
13.	What is Eclat algorithm?	I	1	1	PO1
14.	Write the applications of Association Rule Learning.	I	2	1	PO2
15.	Name the tasks involved in Data Mining.	I	1	1	PO1,PO2
16.	Name the strategies/methods used in Data Transformation.	I	1	1	PO1
17.	Why Data Preprocessing is important?	I	1	1	PO1
18.	Write any 3 advantages of Data Mining.	I	2	1	PO1

19.	What is predictive data mining?	I	1	1	PO3
20.	Write some of the strategies of data reduction.	I	2	1	PO3
21.	Define Classification.	II	1	2	PO3
22.	What are eager learners.	II	1	2	PO3, PO4
23.	What are lazy learners.	II	1	2	PO5, PO6
24.	What is continuous variable.	II	1	2	PO3, PO4
25.	Define Binary Classification.	II	1	2	PO1, PO2
26.	Name some non-linear classification models.	II	1	2	PO4, P05
27.	Name the steps involved in building a classification model.	II	1	2	PO6, PO4
28.	Define feature selection.	II	1	2	PO2, PO3
29.	Define SVM.	II	1	2	PO2, PO1
30.	Define Information Gain.	II	1	2	PO2, PO1
31.	Name some of the advantages of the Decision Tree.	II	1	2	PO4, PO5
32.	What is rule based classification in data mining.	II	1	2	PO3, PO6, PO5
33.	Define Support Vectors.	II	1	2	PO3, PO4
34.	Define Bayesian classification.	II	1	2	PO4, PO5, PO6
35.	Define fuzzy logic.	II	1	2	PO3, PO5
36.	What is defuzzification?	II	1	2	PO4,PO8
37.	Define Fuzzification.	II	1	2	PO4,PO8
38.	Name some applications of Bayesian Classification.	II	1	2	PO4,PO8
39.	How SVM works.	II	1	2	PO3, PO6, PO5
40.	Name the key concepts of SVM.	II	1	2	PO2, PO3
41.	What is prediction?	III	1	3	PO2, PO3
42.	Name the types of prediction in Machine Learning.	III	1	3	PO2, PO3
43.	Define Regression.	III	1	3	PO2, PO3
44.	What is anomaly detection.	III	1	3	PO2, PO3

45.	Name the key components of a Predictive model.	III	1	3	PO2, PO3
46.	Name the steps involved in the Prediction process.	III	1	3	PO2, PO3,PO6
47.	List out some of the challenges in the Prediction process.	III	1	3	PO5,PO6, PO8
48.	Name the Algorithms commonly used for prediction.	III	1	3	PO2, PO3
49.	List out the issues in prediction.	III	1	3	PO2, PO3,PO6
50.	Define ensemble method.	III	1	3	PO5,PO6, PO8
51.	Name the types of ensemble method.	III	1	3	PO3,PO6
52.	Define Bagging.	III	1	3	PO3,PO6
53.	What is boosting.	III	1	3	PO3,PO6
54.	Define Voting Ensembles.	III	1	3	PO3,PO6
55.	Name the types of Regression Analysis.	III	1	3	PO3,PO6
56.	Define Polynomial Regression.	III	1	3	PO3,PO6
57.	Name some applications of regression analysis.	III	1	3	PO1,PO2
58.	Define Classification.	III	1	3	PO1
59.	Name the key concepts in Regression analysis.	III	1	3	PO2
60.	Name some of the advantages of Ensemble Method.	III	1	3	PO2
61.	What is text mining?	IV	1	4	PO2,PO3
62.	Define temporal mining.	IV	1	4	PO1
63.	Define spatial mining.	IV	1	4	PO2,PO3
64.	Name the categories of multimedia data mining.	IV	1	4	PO2
65.	Define web mining.	IV	1	4	PO2,PO3
66.	What is web content mining?	IV	1	4	PO1
67.	Define web usage mining.	IV	1	4	PO1,PO2
68.	Define web structured mining.	IV	1	4	PO2,PO3
69.	What is information retrieval?	IV	1	4	PO2,PO3
70.	Define information extraction.	IV	1	4	PO1,PO2

71.	Define Natural Language Processing.	IV	1	4	PO2,PO3
72.	Why is Text Mining Important?	IV	1	4	PO2,PO3
73.	Name some applications of web mining.	IV	1	4	PO5,PO6
74.	Name some common methods for analyzing text mining.	IV	1	4	PO5,PO6, PO7
75.	Name the components of multimedia mining architecture.	IV	1	4	PO2,PO3
76.	Name any three advantages of text mining.	IV	1	4	PO5,PO6
77.	Name any three applications of web mining.	IV	1	4	PO5,PO6, PO7
78.	Name any types of spatial data mining.	IV	1	4	PO2,PO3
79.	Specify the tasks of temporal data mining.	IV	3	4	PO5,PO6
80.	Name some disadvantages of text mining.	IV	1	4	PO5,PO6, PO7
81.	Define Data warehousing.	V	1	5	PO1
82.	Name the characteristics of data warehousing.	V	1	5	PO1,PO2
83.	Define ETL.	V	1	5	PO1
84.	Name the steps to build the data warehousing.	V	1	5	PO2
85.	List out the operations in OLAP.	V	4	5	PO2
86.	What is data mart.	V	1	5	PO1
87.	Define metadata.	V	1	5	PO1
88.	Define OLAP.	V	1	5	PO1
89.	What is MOLAP.	V	1	5	PO1
90.	What is WOLAP.	V	1	5	PO1
91.	Define SOLAP.	V	1	5	PO1
92.	Name the features of OLAP.	V	1	5	PO1
93.	What is Roll-up.	V	1	5	PO1,PO3
94.	Define Drill-Down.	V	1	5	PO1
95.	Define Staging area.	V	1	5	PO1,PO3
96.	Name some of the disadvantages of OLAP.	V	1	5	PO1,PO3

97.	Name some of the Characteristics of OLAP.	V	1	5	PO1
98.	Name some of the features of OLAP.	V	1	5	PO2,PO3
99.	List out Data Warehouse Users.	V	4	5	PO2,PO3
100.	Name some of the data sources from where we get data.	V	1	5	PO2

SECTION – B (8 MARKS)

S.No	Questions	Unit	K Level	CO	PO
101.	Explain Data Mining and why it is important?	I	2	1	PO4,PO5
102.	Explain briefly about the steps involved in Data Mining.	I	2	1	PO5,PO6
103.	Briefly explain Data Preprocessing.	I	2	1	PO6,PO7
104.	Explain the various methods in Data Cleaning.	I	2	1	PO4,PO5
105.	Explain Data Transformation in detail.	I	2	1	PO4,PO5
106.	Explain the tasks of Data Mining.	I	2	1	PO5,PO6
107.	Explain the Functionalities of Data Mining.	I	2	1	PO6,PO7
108.	Explain Association Rule Learning.	I	2	1	PO4,PO5
109.	Explain how does Association Rule Learning work.	I	2	1	PO5,PO7
110.	Describe the applications of Association Rule Mining.	I	2	1	PO6,PO8
111.	Explain the types of Association rule learning.	I	2	1	PO8,PO9
112.	Explain the performance of Data mining in various databases.	I	2	1	PO6,PO10
113.	Explain the metrics used in A-R leaning.	I	2	1	PO4,PO5
114.	Explain the categories of data mining.	I	2	1	PO5,PO6
115.	Identify the advantages and disadvantages of data mining.	I	3	1	PO6,PO7
116.	Define KDD. Explain briefly about the steps involved in KDD.	I	2	1	PO7,PO8
117.	Briefly explain Learners in Classification Algorithm.	II	2	2	PO7,PO8
118.	Write short notes on Classification Types.	II	2	2	PO6,PO7
119.	Explain the types of classifiers algorithm.	II	2	2	PO8,PO4

120.	Explain the steps involved in building a classification model.	II	2	2	PO4,PO5
121.	Write a short note on Evaluation metrics for classification model.	II	2	2	PO4,PO5
122.	Briefly explain various techniques for feature selection.	II	2	2	PO5,PO6
123.	Explain how does classification work.	II	2	2	PO7,PO8
124.	Identify the advantages and disadvantages of decision tree.	II	3	2	PO7,PO9
125.	Explain Rule based Classification in detail.	II	2	2	PO6,PO8
126.	Write a short note on Bayes' theorem.	II	2	2	PO6,PO7
127.	Explain the Architecture of Fuzzy Logic System.	II	2	2	PO8,PO6
128.	Explain fuzzy logic techniques.	II	2	2	PO4,PO6
129.	Explain Backpropagation Algorithm briefly.	II	2	2	PO5,PO6
130.	Write a short note on the advantages and disadvantages of support vector machine.	II	2	2	PO6,PO7
131.	Write a short note on linear and non-linear SVM.	II	2	2	PO7,PO8
132.	Write a short note on linear and non-linear classifiers.	II	2	2	PO7,PO8
133.	Write a short note on key components of Predictive Model.	III	2	3	PO6,PO9
134.	Explain the steps involved in predictive process.	III	2	3	PO8,PO4
135.	Write a note on application of prediction.	III	2	3	PO4,PO5
136.	Demonstrate the issues while preparing the data for prediction.	III	2	3	PO5,PO6
137.	Explain the fundamental metrics used to access the model's prediction.	III	2	3	PO6,PO7
138.	Write a note on Voting and Stacking ensemble method.	III	2	3	PO7,PO8
139.	Write a note on Bagging and Boosting ensemble method.	III	2	3	PO7,PO8
140.	Explain the advantages and Challenges of ensemble method.	III	2	3	PO9,PO10
141.	What is Regression Analysis? Explain the key components in regression analysis.	III	2	3	PO8,PO9
142.	Explain the type of regression analysis.	III	2	3	PO4,PO5
143.	Write a short note on Applications of Regression analysis.	III	2	3	PO5,PO6
144.	Write a short note on the key concepts in Regression analysis.	III	2	3	PO6,PO7

145.	Explain when to use Ensemble Method?	III	2	3	PO7,PO8
146.	Discuss about Accuracy and Error Measures in Prediction.	III	2	3	PO7,PO8
147.	Write a short note on Prediction Issues.	III	2	3	PO6,PO7
148.	Write a short note on the challenges in Prediction.	III	2	3	PO8,PO10
149.	Explain the architecture of multimedia data mining.	IV	2	4	PO4,PO5
150.	Briefly discuss the advantages of web mining.	IV	2	4	PO5,PO6
151.	Briefly discuss the applications of web mining.	IV	2	4	PO6,PO7
152.	Explain the types of web mining.	IV	2	4	PO7,PO8
153.	Briefly discuss the Text Mining Techniques.	IV	2	4	PO7,PO8
154.	Explain the advantages of text mining.	IV	2	4	PO6,PO7
155.	Define text mining and explain why it is important.	IV	2	4	PO8,PO4
156.	Explain categories of Multimedia Data Mining.	IV	2	4	PO3,PO5,PO7
157.	Differentiate Spatial and Temporal data mining.	IV	2	4	PO6,PO7,PO8
158.	Explain the types of Multimedia Applications.	IV	2	4	PO4,PO5
159.	Briefly discuss the advantages of temporal data mining.	IV	2	4	PO5,PO6
160.	Briefly explain Spatial data mining tasks.	IV	2	4	PO6,PO8
161.	Explain Spatial data mining in detail.	IV	2	4	PO7,PO9
162.	Explain the disadvantages of text mining.	IV	2	4	PO7,PO10
163.	Explain the application of multimedia mining.	IV	2	4	PO6,PO7
164.	Explain the common methods for analyzing Text Mining.	IV	2	4	PO8,PO4
165.	Explain what is data warehousing and mention the characteristics of data warehousing.	V	2	5	PO4,PO5
166.	Briefly explain the advantages of OLAP.	V	2	5	PO5,PO6
167.	Explain the list of operations in OLAP.	V	2	5	PO6,PO7
168.	Briefly explain the top-down architecture of data warehousing.	V	2	5	PO7,PO9
169.	Explain any three types of OLAP.	V	2	5	PO7,PO8
170.	Write short notes on the characteristics of OLAP.	V	2	5	PO6,PO9

171.	Explain the benefits of OLAP.	V	2	5	PO8,PO4
172.	Write a short note on metadata and data mart.	V	2	5	PO3,PO5,PO6
173.	Briefly explain the Advantages of Top-Down Approach.	V	2	5	PO11,PO1
174.	Briefly explain the bottom-up architecture of data warehousing.	V	2	5	PO11,PO1
175.	Briefly explain the disadvantages of OLAP.	V	2	5	PO11,PO1
176.	Explain the need for OLAP.	V	2	5	PO11,PO3
177.	Write a short note on Roll-up and Drill-down.	V	2	5	PO8,PO4
178.	Discuss about the advantages of Bottom-up Approach.	V	2	5	PO3,PO5,PO6
179.	Discuss about the advantages of Top-down approach.	V	2	5	PO11,PO1
180.	Discuss about any four key components of Data Warehousing.	V	2	5	PO4,PO5,PO6

SECTION – C (15 MARKS)

S. No	Questions	Unit	K Level	CO	PO
181.	Explain the steps in Data Preprocessing.	I	2	1	PO4,PO6, PO7
182.	Explain Association Rule Learning with algorithms.	I	22	1	PO5,PO6
183.	Explain the tasks and functionalities in Data Mining.	I	2	1	PO4,PO5
184.	Explain the Association Rule metrics to find the association between data items.	I	2	1	PO7,PO8
185.	Discuss the advantages and disadvantages of Data Mining.	I	2	1	PO4,PO5
186.	Explain the steps involved in the data mining process.	I	2	1	PO3,PO5
187.	Write about the application of data mining in various domains.	I	2	1	PO6,PO7
188.	Explain all the metrics used in A-R learning to find the relationship with examples.	I	2	1	PO4,PO6
189.	Describe decision tree algorithm and explain how it works.	II	3	2	PO3,PO4
190.	Explain in detail about back propagation classification.	II	2	2	PO6,PO7
191.	Explain in detail about Fuzzy Logic classification techniques.	II	2	2	PO4,PO6
192.	Explain in detail about Support Vector Machine.	II	2	2	PO3,PO4
193.	Explain in detail about Bayesian Classification.	II	2	2	PO6,PO7
194.	Describe rule-based classification with example.	II	2	2	PO4,PO6

195.	Explain classification types in machine learning in detail.	II	2	2	PO3,PO4
196.	Define classification and explain evaluation metrics for classification models in Machine Learning.	II	2	2	PO6,PO7
197.	Explain in detail about prediction and its types.	III	2	3	PO4,PO3
198.	Explain the steps involved in prediction process and the challenges in it.	III	2	3	PO3,PO4
199.	Explain the types of ensemble methods.	III	2	3	PO4,PO5
200.	Explain regression analysis in detail.	III	2	3	PO3,PO6
201.	Define Prediction and explain the commonly used algorithms for prediction.	III	2	3	PO6,PO7
202.	Illustrate the challenges and issues in Prediction.	III	3	3	PO4,PO3
203.	Describe regression analysis with Key Concepts in Regression Analysis.	III	2	3	PO3,PO4
204.	Describe the types and steps in the Prediction Process.	III	2	3	PO4,PO5
205.	Illustrate Temporal data mining in detail.	IV	3	4	PO3,PO4
206.	Explain in detail about web mining.	IV	2	4	PO6,PO7
207.	Describe text mining with its techniques in detail.	IV	3	4	PO4,PO5
208.	Explain in detail about Multimedia Data Mining.	IV	2	4	PO3,PO4
209.	Explain in detail about types and categories of multimedia data mining.	IV	2	4	PO4,PO5
210.	Describe Spatial data mining in detail.	IV	2	4	PO3,PO6
211.	Illustrate the advantages and disadvantages of temporal data mining.	IV	3	4	PO6,PO7
212.	Explain the challenges with Multimedia database.	IV	2	4	PO4,PO3
213.	Explain the types of architecture of data warehousing.	V	2	5	PO3,PO4
214.	Explain how to build a data warehouse in detail.	V	2	5	PO4,PO5
215.	Explain OLAP and its types in detail.	V	2	5	PO3,PO4
216.	Explain the benefits and characteristics of OLAP.	V	2	5	PO6,PO7
217.	Explain OLAP operations with its advantages and disadvantages of OLAP.	V	2	5	PO4,PO3
218.	Describe about top-down architecture of data warehousing with its advantages.	V	2	5	PO3,PO4
219.	Describe bottom-up architecture of data warehousing with its advantages.	V	2	5	PO4,PO5
220.	Define data warehouse and describe the key components of data warehouse.	V	2	5	PO3,PO6

K1- Remember; K2- Understand; K3- Apply; K4- Analyze;

K5-Evaluate; K6-Create